Legal

A retiming lag assignment r is legal only if every edge keeps a non-negative register count

The idea

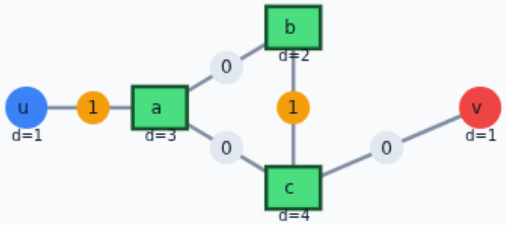
- Check w_r = w + r(dst) − r(src) for e0…e4; one negative edge breaks legality

All w_r ≥ 0

LEGAL RETIME

All w_r ≥ 0

Step 1 / 5 · rule · module02-07-legal-retime



Explanation

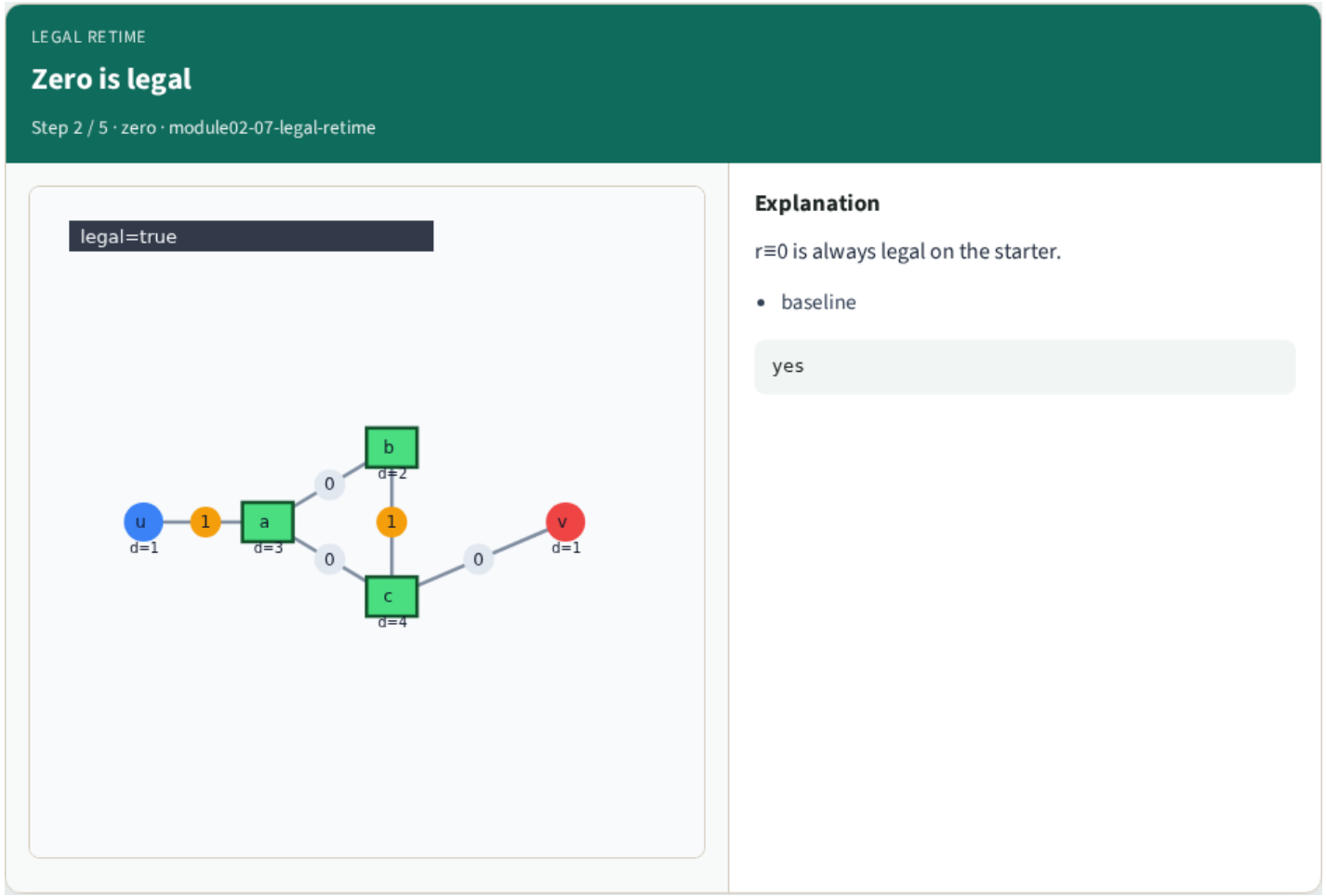
A retiming is legal iff every edge keeps non-negative weight.

- constraint

```
hard
```

All w_r ≥ 0

Zero is legal



Zero is legal

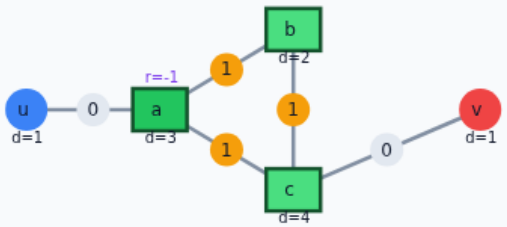
Forward a legal

LEGAL RETIME

Forward a legal

Step 3 / 5 · fwd · module02-07-legal-retime

P=5 regs=3



Explanation

Forward on a stays legal and improves period.

- ok

P=5

Forward a legal

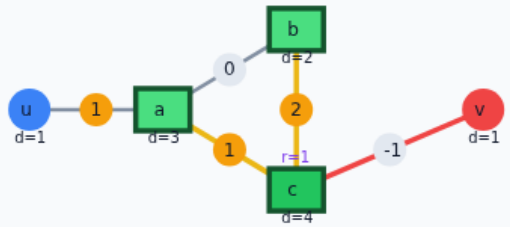
Backward c illegal

LEGAL RETIME

Backward c illegal

Step 4 / 5 · bwd · module02-07-legal-retime

legal=false P=5



Explanation

Backward on c alone violates e3.

- fail

```
w=-1
```

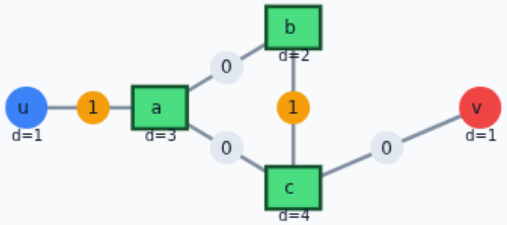
Backward c illegal

Always check

LEGAL RETIME

Always check

Step 5 / 5 · check · module02-07-legal-retime



Explanation

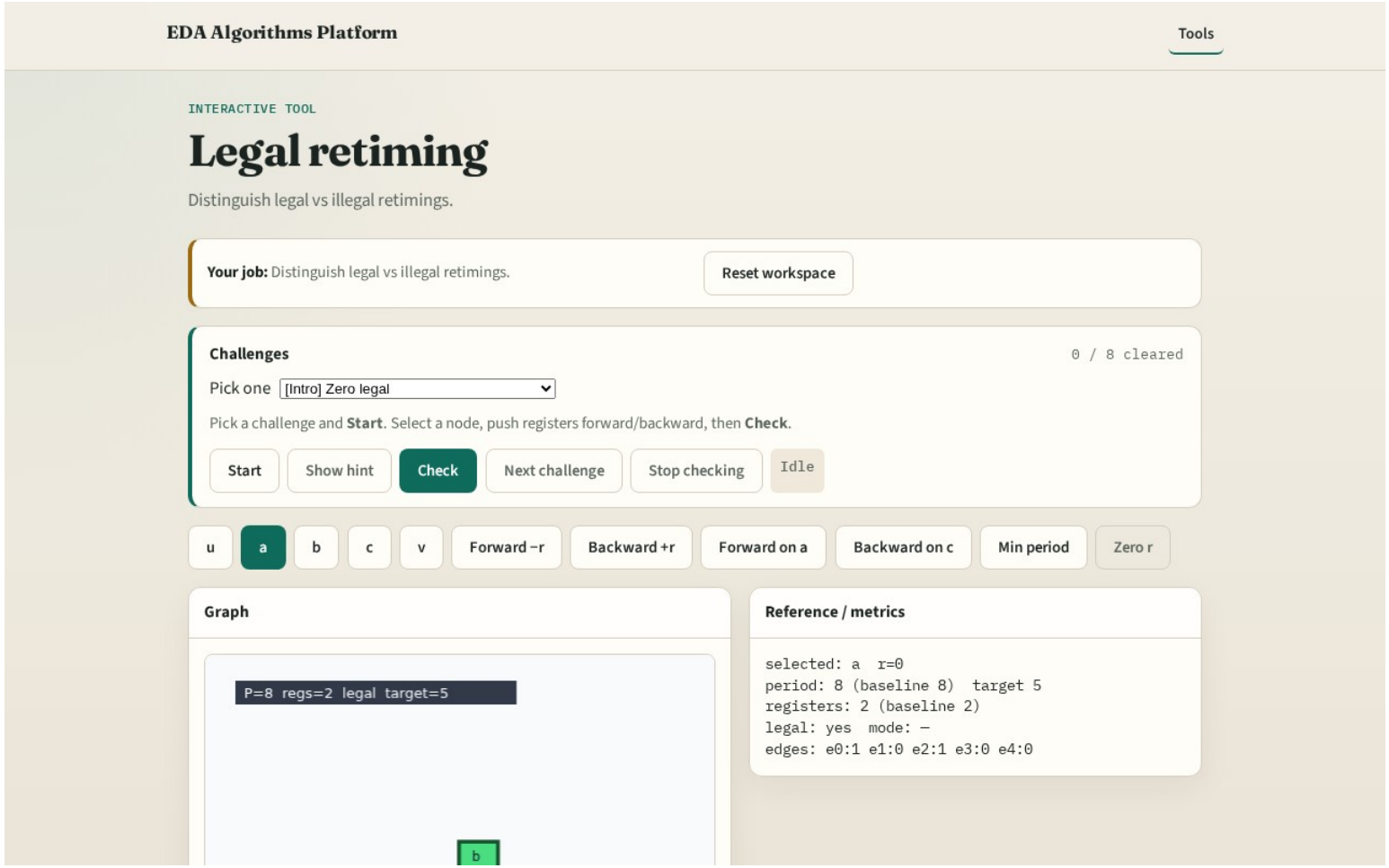
Before trusting period, assert legality.

- is_legal

```
gate
```

Always check

Browser lab track



Browser lab starter

Implement track

- Test is_legal after forward on a (True) and backward on c (False)

Pitfalls

- Checking only one edge
- Allowing fractional or negative w in the UI

Your turn

- Next: period bound / FEAS lite